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The Impact of Income Distribution on the Current Account

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Abstract:

This paper analyzes the relationship between inequality and the current account, addressing the role of the functional distribution of income. Using panel data for 60 countries over the period 1975–2011, our results confirm that an increase in the wage share is associated with a decrease in the current account. We also analyze the effects of wage share on saving and investment, maintaining the same control variables. We find that the wage share is negatively correlated with saving and does not have a significant effect on investment. This result is consistent with the theories that connect higher wages with greater consumption and less saving. The relationship is stronger for developing economies, highlighting the structural differences between different groups of countries. Specifically, the relevance and sign of control variables like financial intermediation, fiscal balance, demographic ratios, capital account openness and growth expectations show important differences according to the level of development.

Keywords: aggregate demand, current account, functional distribution of income, inequality

JEL classification: C23, D31, D33, E21, F32, F41

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1 Introduction

Since the origin of economics as an independent subject of study, income distribution has always been a central issue in economic theory and policy. However, interest in distribution problems has increased significantly in the last decades, for several reasons. In the first place, income inequality, measured through different indicators, has increased in almost all advanced economies and also in most of the emerging and developing economies. This phenomenon, which is important in itself, is also connected with various micro- and macroeconomic issues that influence growth and development (Stiglitz 2013; OECD 2014).

Another reason inequality has gained traction in the economic debate is that many authors have seen it – along with global current account imbalances and financial deregulation – as an underlying cause of the international financial crisis that began in 2008 (Rajan 2010). There is also a complementary discussion happening, in academia, international organizations, and at international fora like the G20, about the role of income distribution in global growth prospects.¹ In fact, theories about secular stagnation that remark the lack of global effective demand place a strong emphasis on inequality (Palley 2015).

All these discussions and research inquiries raise the question of inequality's macroeconomic role in any particular country, and its interactions with the global economy. In this paper, we tackle one aspect of the relationship between imbalances and inequality, by investigating how different income distribution configurations, interacting with other domestic macroeconomic variables, affect the rest of the world through the country's current account.

Our goal is to test whether income distribution has any impact on the current account and how this impact works through saving and investment. The empirical strategy is to expand the previous research of Chinn and Prasad (2003), Chinn and Ito (2007), Chinn, Eichengreen, and Ito (2012), and Phillips et al. (2014) none of which incorporate income distribution and inequality variables among the traditional determinants of the current account.

There are different plausible strategies for measuring income inequality. In our analysis, we use the functional income distribution to measure inequality, where data are available for a larger number of countries. In doing this, we are choosing a different approach than the one utilized in recent literature of the financial crisis, which focuses on the well-known fact that those at the high end of the income distribution are capturing a larger share of total income. Without ignoring the contributions made by the empirical literature based on

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the inclusion of top incomes, we note that this method of measuring inequality faces some limitations when the results have to be extrapolated for emerging and developing countries. First, the phenomenon of so-called “super wages” – like CEO compensation and huge executive bonuses – does not seem to be so relevant outside the main global financial centers. Second, the lack of comparability, or the absence of tax information, places considerable limits on the number of countries that could be included in such research. Using the functional distribution of income helps to solve all these issues and allows us to include a larger number of countries in our analysis.

Our hypothesis in this paper – that an increase in the wage share of income results in a decrease of the current account – is consistent with most of the Keynesian tradition and particularly with Kaleckian theory: workers consume almost everything they earn, while capitalists consume only a fraction of their profits (Kalecki 1971). This is equivalent to the idea that, given wage earners’ lower marginal propensity to save, an income distribution in their favor results in an increase in the consumption level. Even though this mechanism has been complicated by the fact that, through the financial sector, consumption can be fueled by debt instead of income, we will argue that it is still predominant, at least in most countries. Thus, if the increase in the wage share implies an increase in aggregate consumption that is not fully compensated by a decrease in investment, then a higher wage share implies a deterioration of the current account.²

There is a long tradition among post-Keynesian authors to consider the impact of the functional distribution of income on aggregate demand. In particular, there is a wide discussion about how growth can be wage-led or profit-led (Hein and Vogel 2008; Stockhammer, Onaran, and Ederer 2009; Blecker 2016) and about the interaction of aggregate demand with the accelerator principle (Blecker 1999; Bortz 2016). This literature provides relevant theoretical and empirical insights regarding how the wage share is related to the main components of the current account (i.e. saving and investment) and, more specifically, with the individual components of aggregate demand. In particular, this literature analyzes the relationship between the wage share and the trade balance. The present work extends the analysis to the current account.

The paper is organized as follows. Section 2 reviews the literature of the determinants of the current account, where the income distribution has been almost absent until some recent works related to the global financial crisis. Section 3 is devoted to the analysis of the wage share and its relationship with indicators of personal income distribution. In Section 4 we analyze how the wage share is related to the current account through the individual components of aggregate demand. In Section 5, we introduce the econometric strategy to test our main hypothesis, that the functional distribution of income is a highly significant determinant of the current account balance. The results are shown in Section 6, and the main conclusions are presented in Section 7. The paper also includes several appendices with robustness checks and details of our empirical work.

2 The Determinants of the Current Account. Is Inequality One of Them?

There are different empirical models to identify the determinants of the current account. In particular, we are interested in the ones that analyze the determinants of saving and investment. For example, in a seminal paper in this literature, Chinn and Prasad (2003) ran panel and cross-section regressions that included as explanatory variables the budget balance, the net foreign assets, the relative income, demographics dependency ratios, a measure of financial deepening and indicators of terms of trade, openness and capital controls. This set of variables has remained almost unchanged in more recent works (Chinn and Ito 2007; Chinn, Eichengreen, and Ito 2012), except with more emphasis on institutional arrangements and in the fiscal balance, in order to test the “saving glut” vs. the “twin deficits” hypothesis. The IMF continued this line of research and adopted it in its External Balance Assessment as the standard methodology for estimating the current account in a multilaterally consistent manner (Phillips et al. 2014).

As we can see, the income distribution is not considered as a determinant of the current account in this literature. This has changed recently, after some economists began to assert that inequality was one of the causes of the global financial crisis. Rajan (2010), one of the most widely acknowledged authors proposing this idea, argued that the crisis originated from the social pressures caused by increasing inequality in the U.S. According to him, the government confronted this situation by promoting mortgages instead of seeking long-term solutions, which would have entailed higher investment in education. In a broader sense, the main question is related to whether better-functioning democratic institutions would have prevented higher inequality from unleashing a financial bubble (Stiglitz 2012; Palley 2015).

Following Rajan’s work, several other authors have analyzed the role of inequality in the financial crisis. It has been suggested that with the worsening of income distribution in the years leading up to the financial crisis, middle- and low-income sectors reacted by getting into debt to maintain their consumption levels. Consequently, the evolution of a long trend of increasing inequality has a correlation with a period of increasing

indebtedness, which became unsustainable over time and unleashed the subprime crisis. The counterpart of this higher indebtedness is a higher aggregate consumption level, lower aggregate savings, and – without an offsetting reaction of investment – a lower current account balance. This line of thought suggests that there is a negative relationship between inequality and the current account balance, and there are some empirical works that support this idea (Al-Hussami and Remesal 2012; Kumhof et al. 2012; Behringer and van Treeck 2013).

Our view is that the main contribution of these empirical works is to incorporate income distribution as a determinant of the current account. In this sense, our paper continues this line of research. However, we are not fully convinced by the suggested negative relationship between inequality and the current account. It is a suggestion that in fact faces certain difficulties at both the theoretical and empirical levels.

In the field of economic theory, the assumption that the financial system allows the decoupling of saving/consumption decisions from current or future income poses a challenge to consumption theories based on permanent income. Several authors have proposed alternative hypotheses to that of permanent income, incorporating psychological elements suggested by behavioral economics. For example, Barba and Pivetti (2009) argue that individual consumption is related to previously acquired consumption levels; therefore, an individual who loses income does not adjust his/her consumption level. Likewise, theories of relational consumption (Duesenberry 1949; Schor 2005) argue that an individual's consumption depends on the consumption level of his peers. These theories are the basis of so-called *expenditure cascades* (Frank 2005; Frank and Levine 2007), which take place when the income captured by top fractiles increases. When the consumption level of the high-income sectors goes up, the middle- and low-income households try to replicate this behavior, regardless of the fact that their incomes may not have changed, or may have even decreased – as was the case for most American households since the 1990s.

There are formal models of how expenditure cascades allow higher inequality to translate into higher indebtedness and a lower current account balance, such as two recent studies by the IMF (Kumhof and Rancière 2010; Kumhof et al. 2012). These models incorporate Keynesian elements when they explain that a worsening of income distribution originates in a weakening of workers' bargaining power. However, these models include the idea that, since workers can borrow, they can sustain or even increase consumption. We think that such a model can be valid in advanced economies with a well-developed financial system. However, in emerging³ economies, income is still a restriction for consumption and is not interchangeable with debt. Furthermore, Stockhammer and Wildauer (2016) do not find evidence of the expenditure cascades in their review of 18 OECD countries.

There are also some empirical difficulties with the evidence that the literature suggests supports a negative relationship between the income captured by top percentiles and the current account balance. First, these studies were based on a small sample of countries (between 18 and 22), most of them advanced economies, because the authors' basic argument was strongly based on the hypothesis of expenditure cascades. For studies based on this theory, top income share is preferable as an inequality indicator to the Gini coefficient, given the insensitivity of the latter to changes in the tails of the distribution. Thus, the sample of countries was limited to the data available in "The World Top Incomes Database," prepared by, among others, Atkinson, Piketty, Saez, and Alvaredo. Even though this database has been expanded with the inclusion of some additional countries, the sample is still limited. Carrera, Rodríguez, and Sardi (2015) tested the hypothesis with a higher number of countries (29) and find that the negative relationship between top incomes and current account balance was not robust.

Second, top income indicators are built on the basis of tax information, meaning that they are only comparable when the legal definitions of taxable income are somewhat consistent and when the tax systems are sufficiently efficient (Galbraith 2014; Amarante 2015). They are based on income declarations, and in several countries only a small proportion of the population makes these declarations. Tax evasion and avoidance are major problems in some emerging countries, and the population that declares incomes is often not representative of the whole population, as it is less than 10% or even 5% of the active population, and at that, may not include the richest individuals. Another major problem is that top incomes are a ratio in which the numerator and denominator come from different sources (income declarations and national accounts). Methodological changes or revisions in the national accounts, which are common in emerging countries, modify the ratio, and it is not always possible to reconstruct the whole series.

The decision to focus in this paper on the functional distribution of income, rather than on the top income share, not only allows us to considerably increase the number of countries in the sample, but also helps us to avoid focusing the theoretical discussion on arguments for which the empirical evidence is still unclear, especially for countries other than the United States. Indeed, assuming that wages have a higher weight in the income of middle- and low-income sectors, a higher wage share should correlate with a decrease in inequality measured by other indicators as Gini or top incomes. This means that analyzing this correlation between inequality indicators is crucial for comparing our results to the ones of the previous literature. This analysis is carried on in the next section.

3 The Wage Share and its Relation with Personal Distribution of Income

The functional distribution of income refers to the division of national income between capital and labor, while the wage share is the part of national income that is devoted to labor compensation (ILO, IMF and OECD 2015). The measure of this variable has some methodological issues related to what is included as labor income. Gollin (2002) explains that for a number of reasons, the labor income of the self-employed is often treated incorrectly as capital income, when it could be labor income or a mix of both. One approach to deal with this problem is to adjust the wage share estimated from national accounts by self-employment rates. An alternative is the one proposed by Karabarbounis and Neiman (2013), who constructed a dataset of wage share in the income of 112 countries over the period 1975–2012, based in the corporate sector. As the authors explain, the labor share measured within the corporate sector is not affected by the statistical imputation of wages from the combined capital and labor income earned by sole proprietors and unincorporated enterprises, which Gollin (2002) highlighted as problematic for the consistent measurement of the labor share. In our opinion, Karabarbounis and Neiman's database is the largest available with standardized information on the functional distribution of income, and it is thus the one we use in this paper.

Movements in the wage share are often interpreted as changes in real wages. However, as Saint-Paul and Bentolila (2003) note, this assumption can be mistaken: some countries have experienced a significant decline in the wage share without a decline in real wages. In their study of 13 industries in 12 OECD countries, they find evidence in favor of an empirical relationship between the labor share and the capital-output ratio, the capital-augmenting technical progress, and with factors that create a wedge between the shadow marginal cost of labor and the wage, like labor adjustment costs and, marginally, union bargaining power. Similarly, Checchi and García-Peñalosa (2005) find that labor market institutions like union density, the minimum wage and the unemployment rates have an impact on the wage share.

For a long time, the empirical evidence suggested that there was stability in the wage share in income, at least for long periods, to such an extent that this was one of the well-known *stylized facts* given by Kaldor (1961). In his studies of economic growth, he had found that the functional distribution of income remained stable over the years, and this translated into an increase in real wages proportional to the average productivity growth rate. However, recent studies have indicated that the current stylized fact is that the capital income share has gone up rather than staying constant (Rodríguez and Jayadev 2010; Kanbur and Stiglitz 2015). In addition, the aforementioned work by Karabarbounis and Neiman (2013) argues that – at least since the beginning of the 1980s – there has been a decreasing global trend of wage share in income, which may be explained by a fall in the relative price of investment goods. Consequently, the efficiency gains of the productive sectors, usually attributed to progress in information technology (IT), lead to a more intensive use of capital, displacing human work in relative terms.

One question that is relevant to this paper is whether the recent decline of the wage share is in some way related with the increase in inequality in personal income. Checchi and García-Peñalosa (2005) model the Gini coefficient as a function of population proportions, the number of capital owners, the wage share, the wage differential, and the unemployment subsidy. They argue that a lower wage share will increase inequality by raising profits and thus the income of the richest individuals, and they find evidence that the trend reversal in income inequality observed in most European countries at the end of the 1970s is largely explained by the contemporaneous decline in the wage share.

This is highly relevant for the interpretation of our results. If a decrease in the wage share can be linked to higher inequality in personal incomes, our hypothesis goes in the opposite direction of the results of Al-Hussami and Remesal (2012), Behringer and van Treeck (2013), and Kumhof et al. (2012). That is, higher inequality is related with an improvement, not a decrease, in the current account. In order to test this, we were able to analyze the correlation between the wage share and income captured by the top 1% for 24 countries,⁴ while an analysis of the correlation using the Gini coefficient⁵ could be extended to 60 countries, which we eventually included in our empirical work. For the total sample, a negative correlation is observed, even though it is not very significant. This is illustrated in Figure 1.

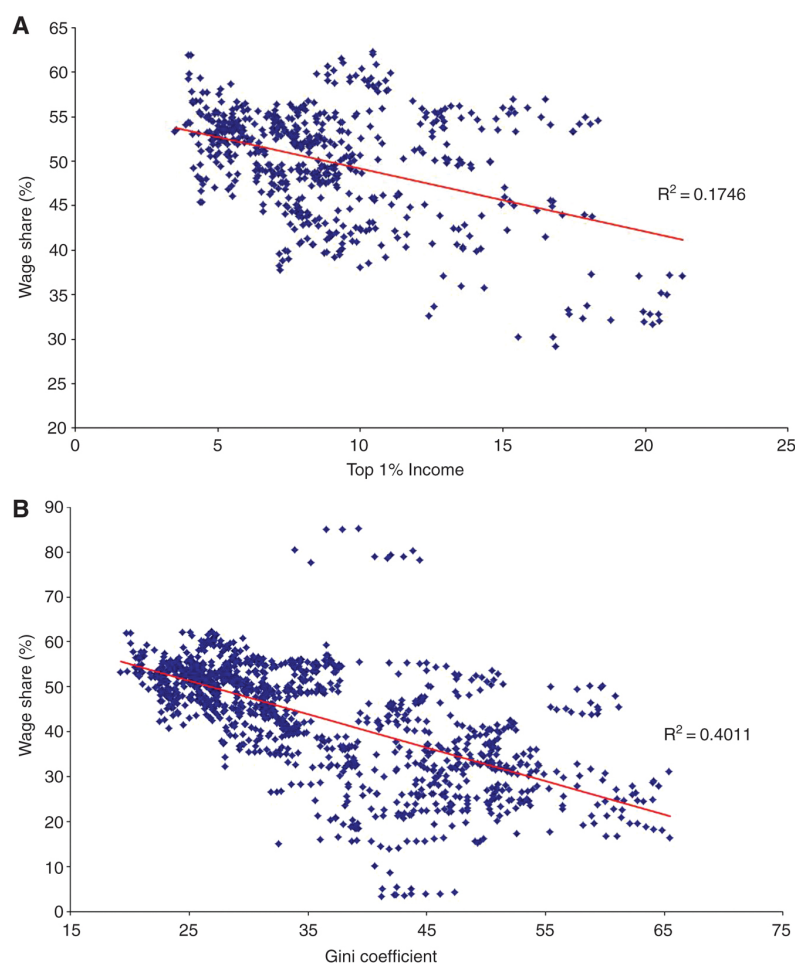


Figure 1: A. Dispersion between Wage Share and Top 1 (24 countries). B. Dispersion between Wage Share and Gini (60 countries)

Moreover, this negative correlation for the total sample is in fact relative as the situation may differ considerably between countries. Even though negative correlations prevail, a case-by-case analysis reveals that there are countries for which the correlation is null or even positive, as illustrated in Table 1. Appendix A describes the estimated correlations for each country under analysis, while Appendix B shows that these correlations remain stable both when contemporary values of each indicator are considered and when a lag or lead in any of the variables is included.

Table 1: Different Correlations Between Wage Share, Top 1%, and Gini.

	Wage share/Top 1%	Wage share/Gini
Countries with correlations < -0.5	12	20
Countries with correlations of -0.5 to 0	7	21
Countries with correlations of $0-0.5$	3	6
Countries with correlations > 0.5	2	13

Source: Authors' calculations using data series from Karabarbounis and Neiman (2013), the World Top Incomes Database, and The Standardized World Income Inequality Database.

This analysis shows that, in general, a larger share of wages is related to an improvement in income distribution, measured either by the Gini coefficient or the top 1%. However, we must be extremely careful with this interpretation. A more extensive analysis of how different inequality indicators are related is needed. In particular, we need to analyze the role of the “super wages” in the upper tails of the distribution, which seems to be relevant in some advanced economies.

4 The Role of the Wage Share in the Main Components of the Aggregate Demand

After describing the recent trends of the wage share and its relation with personal income distribution, it is useful to revise the relationship between the wage share and the main components of the current account. In this regard, we need to remember that there are two ways of seeing the current account: (1) as an excess of savings over investment, and (2) as the sum of the trade balance, net factor income and net transfers, where the trade balance usually prevails. As a result, and considering that savings decisions are inversely related with consumption decisions, the analysis of the components of the current account is closely related with the analysis of the individual components of aggregate demand.

The most consistent studies regarding how the wage share relates with the individual components of aggregate demand came from post-Keynesian literature. As we mentioned before, there is a wide discussion about how growth can be wage-led or profit-led that is beyond the scopes of this paper. However, the empirical literature that follows a “structural” approach – that is, the approach that analyzes the relationship of wage share and the individual components of aggregate demand – is highly relevant to our work.

The most robust result of these studies is that the wage share has a positive effect on consumption (Hein and Vogel 2008; Stockhammer, Onaran, and Ederer 2009; Blecker 2016), as a result of the higher marginal propensity to consume (MPC) that labor income induces as compared to capital income. While workers typically spend all or most of their wage income on current consumption, a portion of profits is retained by firms to finance investment, acquisitions or other purposes. According to Blecker (2016), given that most consumers are reliant primarily on labor income, the MPC that wage income induces is also likely to be higher in the long run than in the short run. In contrast, the consumption of upper-class households with very high saving rates is not closely tied to their income over any particular time horizon. Nevertheless, this conclusion might not hold in countries where consumers have less access to credit, so that their consumption expenditures are more restricted by their disposable income even in the short run (as happens in developing or emerging countries).

Less clear is the expected relationship between wage share and total investment. While business investment will depend negatively on the wage share, residential investment may be neutral or even react positively to changes in the wage share through homes construction (Stockhammer and Wildauer 2016). This means that when there is no distinction between business and residential investment, the wage share could appear to be insignificant to investment, or have either positive or negative signs. Even when the distinction between the two different types of investment is possible, the negative effect of wage share on business investment could be largely short-run (Blecker 2016) or change over time due to the effect on investment flows of globalization (Stockhammer, Hein, and Grafl 2011).

Finally, the literature finds a negative relationship between the wage share and the trade balance (Stockhammer, Onaran, and Ederer 2009). Rearranging the definition of the wage share (w) we can reach the following expression:

$$w = \frac{WL}{PY} = \frac{(W/P)}{(Y/L)}$$

where W is the nominal wage rate, L is labor employed, P is the price level, and Y is real output. As we can see in the expression on the right, the wage share can be seen as the ratio of the real wage to labor productivity. As such, the wage share is closely related to the real unit labor cost (Blecker 1989; Stockhammer and Wildauer 2016), implying that an increase in wages results in a loss of the competitiveness of national products. In this way, the wage share is also closely related to the real effective exchange rate (REER). This *cost/exchange rate effect* is the main explanation found in the post-Keynesian literature regarding the negative relationship between the wage share and the trade balance. However, the *income effect* is another channel that can arise when an increase in wages results in an increase in imports, with the same effect in the trade balance as in the cost/exchange rate effect.

As the trade balance is the main component of the current account, it is natural to expect that the relationship between the wage share and the current account is similar to the relationship between the wage share and the trade balance. However, this question needs to be tested empirically, especially after some recent papers have suggested that an increase in personal inequality is associated with a lower current account balance (Al-Hussami and Remesal 2012; Kumhof et al. 2012; Behringer and van Treeck 2013).

5 Methodology and Econometric Strategy

In this section, we describe our strategy for estimating the relationship between the current account and the functional distribution of income. In particular, we will try to show that a higher wage share is associated with a lower current account balance. Moreover, we will analyze how the relation between the wage share and the current account can be decomposed in the investment and saving sides.

In order to achieve this, we run separate regressions using as dependent variables the current account, total investment, and savings,⁶ all of them as ratios of GDP. This is the same procedure followed by the previous literature (Chinn and Prasad 2003; Chinn and Ito 2007; Chinn, Eichengreen, and Ito 2012; Phillips et al. 2014). However, since these papers did not include the wage share as a determinant of the current account, we also need to compare our results with the ones of the post-Keynesian literature described in the previous section.

Additionally, we try to identify whether the negative relationship between the wage share and the current account is due to an income channel, a cost/exchange rate channel or both, running regressions including the REER. However, as Phillips et al. (2014) note, estimating the current account as a function of REER would be inappropriate since both are endogenous and simultaneously determined as a function of the same variables. Considering this, we only include as a robustness check the main regressions of the current account with lags of the REER.

The estimation methodology follows most of the recommendations of IMF technical notes (Phillips et al. 2014) and uses pooled generalized least squares (GLS) with a first-order autoregressive AR(1) correction, given the strong autocorrelation shown by the current account series. We apply this methodology to an unbalanced data panel of 60 countries with annual data on different variables throughout the period 1975–2011. We attempted to identify the cyclical sources of current account behavior and for this reason we discarded the use of 4- or 5-year non-overlapping averages.

Many cross-country panel studies use fixed effects in order to soak up all country-specific effects. Even though we share the opinion of Chinn and Prasad (2003) that including fixed effects would detract from many of the economically meaningful parts of the analysis, as a robustness check we also include the main regression with fixed effects.

In addition to the wage share, we include the usual control variables recommended by the literature of current account determinants. Below is a description of these variables, together with the sign expected for the regressions based on economic theory. Appendix C contains a complete description of the sources of these variables.

Wage Share in Total Income (“wage_share”). We expect a negative sign for the estimated coefficient in the savings regression, due to its positive effect on consumption. As discussed in Section 4, the expected sign in the total investment regression remains uncertain. Overall, we expect a negative sign in the wage share coefficient in the current account regression. As a robustness check, we analyzed the existence of nonlinearities and the inclusion of lags of the wage share in its relationship with the current account.

Gini Coefficient (“gini”). We include the Gini as another control of inequality. Previous literature is ambiguous about its sign. While Carrera, Rodríguez, and Sardi (2015) suggests that it is not a significant variable in determining the current account, Behringer and van Treeck (2013) found a negative relationship.

Net Foreign Assets Position (% GDP) (“nfa”). The variable is included in the regressions with a lag and a positive sign in its coefficient is expected in the current account regression as a higher accumulation of external assets in t generates a rent flow in $t+1$. Previous literature confirms that this variable correlates with higher savings and current account balance, without a significant role for investment (Chinn, Eichengreen, and Ito 2012). According to Phillips et al. (2014), this positive relationship between the lag of the NFA and the current value of the current account may disappear when the NFA adopts very negative values due to the emergence of sustainability problems. Therefore, following his recommendations, we include a dummy (“nfa_60”), which adopts a value of 1 when the NFA is lower than 60% (negative) of GDP and a value of zero in all other cases.

Population Dependence Ratios (“dep_young” and “dep_elderly”). According to the lifecycle hypothesis, a higher proportion of young and elderly population reduces savings, as they are net consumers, thus negatively affecting the current account. However, several factors can induce the elderly to continue saving: the wish to leave an inheritance, or uncertainty about life expectancy and future expenses. For this reason, and to capture these effects properly, two different dependence ratios are used: between 0 and 14 years old, and over 65 years old. For both ratios, the denominator is the population between 15 and 64 years old. Chinn, Eichengreen, and Ito (2012) found that the dependency ratio of the youngest people is negatively related both to saving and investment in every group of countries. The total effect on the current account is negative in the less developed countries, but it has a non-significant effect in the current account of industrial and emerging countries. They found the dependency ratio of the elderly to be negatively related with saving in emerging and industrial countries, but it is only negatively related to investment in industrial countries. For the current account, they found that this ratio is significant and negative for emerging and less developed countries, but not for industrial ones.

Credit to the Private Sector (% GDP) (“cred_tot”). The effects of the size of the financial sector on the current account are also ambiguous. On the one hand, they may increase saving opportunities, leading to a higher account balance. On the other hand, they allow access to financing to increase consumption levels, adversely affecting the current account. Chinn, Eichengreen, and Ito (2012) found that this variable is not significant for the current account but it is positively related to both investment and saving.

Fiscal Balance (% GDP) ("fiscal_balance"). It is expected to be positively related to both saving and investment, but with a weaker effect on the latter, resulting in a positive sign in the current account regression. These results are confirmed by the literature (Chinn, Eichengreen, and Ito 2012; Phillips et al. 2014).

Output Gap (% GDP) ("y_gap"). This variable is included in order to control for the effects of the business cycle on the current account. As a cyclically lower output is typically associated with high saving and lower investment (Phillips et al. 2014) – i.e. a higher current account – we expect a negative sign for the coefficient of this variable. Phillips et al. (2014) found that this variable is positively related to both saving and investment, but in a stronger way for the latter, resulting in a negative sign for the coefficient in the current account regression.

Five-Year Growth Forecast ("5y_gdp_f"). Future growth expectations tend to promote investment, reducing the current account balance; therefore, a negative sign is expected for the coefficient. The growth forecast for the year $t+5$ is considered, rather than the annualized growth forecast for the next 5 years, so as to focus on growth perspectives rather than on cyclical fluctuations. Phillips et al. (2014) found that this variable is positively related to both saving and investment, but in a stronger way for the latter, resulting in a negative sign for the coefficient in the current account regression.

PPP-Converted GDP Per Capita ("rel_pcgdp"). This variable is included as a proxy of productivity or the level of development, and it is taken in terms relative to the GDP per capita of the average of three large economies: the United States, Germany, and Japan. Even though the empirical evidence is contradictory in this respect, capital is expected to flow from countries with excess savings to countries with higher investment opportunities; as a result, the sign of the coefficient should be positive. For this to happen, policies must allow these capital flows. Thus, the variable (with a lag) is included alone and interacted with a capital account openness index ("k_openness"), which is also included alone for a better interpretation of the coefficients. Phillips et al. (2014) found that this variable by itself is not significant either for saving, investment or current account, but the interaction term is positive for the three of them.

Dummy of "Financial Center" ("fin"). Following Phillips et al. (2014), this variable adopts a value equal to one for relatively small economies with developed financial systems. Deciding which countries fall under this category is somewhat arbitrary, but, following the recommendations of the aforementioned authors, we decided to include Belgium, Luxembourg, the Netherlands, Singapore, and Switzerland. A positive value is expected for this variable.

Real Effective Exchange Rate ("reer"). As a robustness check, we included this variable with lags in order to analyze if the relationship between the wage share and the current account is due to an income effect or an exchange rate effect, as it was explained before. An increase in REER implies that exports become more expensive and imports become cheaper; therefore, an increase indicates a loss in trade competitiveness.

Some of these variables, such as the net foreign assets position, are not independently determined in each country, but their values incorporate the common factors in the cross-sectional dimension. The variables that do not meet this criterion were demeaned using GDP for weighting,⁷ as recommended by literature. As a result, the demeaned variables were: the demographic ratios, the GDP growth forecast, the output gap, the credit/GDP ratio, and the fiscal balance.

To test the robustness of our results, five different specifications of the model were run with different panel data compositions: (i) including an indicator of the functional distribution of income; (ii) including an indicator of the personal distribution of income; (iii) including both income distribution indicators (functional and personal); (iv) including a linear term and a squared term of the functional income distribution indicator; (v) including the contemporary value and two lags of the functional income distribution indicator. All these specifications were tested for: (i) the total sample of 60 countries; (ii) a panel of advanced countries and another panel of emerging countries. These different panel compositions were used to identify structural differences between countries. In Appendix D we list the 60 countries included in our sample.

6 Results

Table 2 shows the results for the total sample. It can be observed that the coefficient associated with wage share is highly significant in all cases and has a negative sign. This means that a higher share of wages in total income is associated with a lower current account balance. On the other hand, the Gini coefficient is not significant when it is included together with the functional distribution of income indicator, nor is it significant when the wage indicator is not included. The values of the wage share coefficients are similar in models (1) and (3), and they nearly double in models including squared terms or lags. In model (4), the positive sign of the squared term could indicate that the negative effect exerted by wage share on the current account decreases for high

values of wage share, but the coefficient is not significant. In model (5) the lag terms are both significant and positive, indicating a negative but not explosive relationship between the wage share and the current account.

Table 2: Results for the Total Sample.

	(1) CA	(2) CA	(3) CA	(4) CA	(5) CA
wage_share	−0.229*** (−6.50)		−0.234*** (−6.51)	−0.424*** (−3.26)	−0.494*** (−7.47)
gini		0.025 (0.51)	−0.030 (−0.59)		
wage_share ²				0.221 (1.54)	
wage_share(−1)					0.132* (1.90)
wage_share(−2)					0.182*** (2.84)
y_gap	−0.003*** (−7.16)	−0.001*** (−3.06)	−0.003*** (−7.20)	−0.003*** (−7.13)	−0.003*** (−6.62)
k_openness	−0.006* (−1.86)	−0.006** (−2.09)	−0.007** (−2.02)	−0.005* (−1.66)	−0.007* (−1.92)
rel_pcgdp(−1)	0.023* (1.66)	0.003 (0.19)	0.013 (0.78)	0.024* (1.70)	0.023 (1.63)
rel_pcgdp(−1)*k_openness	0.013** (2.49)	0.016*** (2.79)	0.016*** (2.74)	0.012** (2.31)	0.014*** (2.72)
5y_gdp_f	−0.495*** (−3.47)	−0.389*** (−3.31)	−0.493*** (−3.42)	−0.485*** (−3.41)	−0.092 (−0.58)
nfa(−1)	0.028*** (3.89)	0.020*** (2.68)	0.028*** (3.78)	0.027*** (3.67)	0.029*** (3.95)
nfa_60(−1)	−0.029*** (−3.96)	−0.021*** (−2.87)	−0.029*** (−3.90)	−0.028*** (−3.85)	−0.028*** (−3.84)
dep_young	−0.195*** (−6.47)	−0.102*** (−3.69)	−0.202*** (−6.51)	−0.218*** (−6.44)	−0.178*** (−5.71)
dep_elderly	−0.244*** (−3.06)	−0.187** (−2.28)	−0.264*** (−3.07)	−0.260*** (−3.18)	−0.198** (−2.42)
cred_tot	−0.013* (−1.85)	−0.022*** (−3.19)	−0.012* (−1.66)	−0.013* (−1.81)	−0.013* (−1.80)
fiscal_balance	0.183*** (5.80)	0.157*** (4.95)	0.181*** (5.68)	0.177*** (5.61)	0.181*** (5.72)
fin	0.046*** (3.26)	0.051*** (3.58)	0.046*** (3.25)	0.047*** (3.28)	0.046*** (3.18)
_cons	0.072*** (3.69)	−0.037 (−1.64)	0.088*** (3.04)	0.112*** (3.44)	0.047** (2.31)
Countries	60	60	60	60	60
N	1051	1298	1033	1051	973
RMSE	0.044	0.035	0.044	0.045	0.043

t-Statistics are reported in parentheses. *, ** and *** indicate the rejection of the null hypothesis at the 10%, 5% and 1% statistical significance levels, respectively. RMSE denotes squared root of mean squared error.

The control variables are significant and have the expected signs. A higher stock of foreign assets and a higher fiscal balance are associated with a higher current account balance. On the other hand, a higher output gap and higher future growth expectations have a negative relationship with the current account. Similarly, greater financial depth measured by the ratio of total credit to GDP is associated with a lower current account.

For demographic variables, a higher proportion of a population younger than 15 years or older than 64 years correlates with a lower current account. The dummies are also significant and have the expected signs: financial centers have a higher current account balance, while the positive effect of foreign asset accumulation diminishes when the net foreign assets position becomes very negative.

A different analysis is required for relative GDP per capita and the openness of the capital account, as these variables are interacted in the regressions. The following equations will help to show the interpretation of the related coefficients.

$$CA_{i,t} = \beta_0 + \beta_1 \times rel_pcgdp_{i,t-1} + \beta_2 \times k_openness_{i,t} + \beta_3 \times rel_pcgdp_{i,t-1} \times k_openness_{i,t} + BX_{it}$$

$$\frac{dCA_{i,t}}{drel_pcgdp_{i,t-1}} = \beta_1 + \beta_3 \times k_openness_{i,t} ; \quad \frac{dCA_{i,t}}{dk_openness_{i,t}} = \beta_2 + \beta_3 \times rel_pcgdp_{i,t-1}$$

where CA is the current account, rel_pcgdp is the relative GDP per capita, $k_openness$ is the indicator of the openness of the capital account and X is the vector of other determinants of the current account included in the regressions. The partial derivatives show how the relationship between the current account and one of these variables depends on the other variable. In particular, if we take the results of the model 3 as a reference, the partial derivatives take this form:

$$\frac{dCA_{i,t}}{drel_pcgdp_{i,t-1}} = 0.016 \times k_openness_{i,t} ; \quad \frac{dCA_{i,t}}{dk_openness_{i,t}} = -0.007 + 0.016 \times rel_pcgdp_{i,t-1}$$

As the capital account openness index built by Chinn and Ito (2006) take positive and negative values, the first partial derivative can be of both signs. This means that in closed or less open countries, a higher GDP per capita corresponds with a lower current account, but in more open economies we found the opposite relationship. In the same way, the second partial derivative shows that the effect of the capital account openness on the current account can be either positive or negative, depending on the GDP per capita. In advanced countries, the sign of this relationship is positive, while in countries with lower GDP per capita, the relationship is negative. In our sample, approximately half of the countries have an average GDP per capita where the relationship turns negative.

As we have discussed in the previous sections, the wage share could be related to the current account either through saving or investment. In order to analyze this, we present separate regressions for saving and investment. The results are shown in Table 3 and Table 4. We can see that the wage share is highly significant and with negative sign in the saving regressions, but it is not a significant determinant of investment. However, as we mentioned in Section 4, this could be the result of a negative relationship between the wage share and business investment and a positive one with residential investment.

Table 3: Investment Regression. Total Sample.

	(1) Invest	(2) Invest	(3) Invest	(4) Invest	(5) Invest
wage_share	-0.018 (-0.58)		-0.022 (-0.71)	0.017 (0.15)	0.076 (1.34)
gini		-0.003 (-0.07)	-0.099** (-2.23)		
wage_share ²				-0.040 (-0.32)	
wage_share(-1)					-0.067 (-1.14)
wage_share (-2)					-0.045 (-0.82)
y_gap	0.003*** (8.91)	0.001*** (6.12)	0.003*** (8.90)	0.003*** (8.87)	0.003*** (7.92)
k_openness	-0.001 (-0.44)	-0.001 (-0.41)	-0.001 (-0.43)	-0.001 (-0.50)	-0.000 (-0.05)
rel_pcgdp(-1)	-0.016 (-1.29)	-0.022 (-1.43)	-0.026* (-1.74)	-0.016 (-1.31)	-0.013 (-1.01)
rel_pcgdp(-1)*k_openness	-0.002 (-0.48)	-0.005 (-0.97)	-0.001 (-0.26)	-0.002 (-0.43)	-0.003 (-0.70)
5y_gdp_f	0.799*** (6.52)	0.724*** (7.48)	0.806*** (6.47)	0.804*** (6.55)	0.450*** (3.29)
nfa(-1)	0.002 (0.37)	0.008 (1.33)	0.004 (0.70)	0.003 (0.40)	0.002 (0.31)
nfa_60(-1)	0.014** (2.26)	0.010* (1.75)	0.012* (1.92)	0.014** (2.23)	0.012* (1.82)
dep_young	-0.167*** (-6.15)	-0.095*** (-3.71)	-0.165*** (-6.08)	-0.163*** (-5.48)	-0.191*** (-6.73)
dep_elderly	-0.207*** (-2.89)	-0.105 (-1.38)	-0.262*** (-3.48)	-0.203*** (-2.82)	-0.286*** (-3.86)
cred_tot	0.015** (2.40)	0.022*** (3.48)	0.016** (2.56)	0.015** (2.41)	0.013** (2.00)
fiscal_balance	0.065** (2.39)	0.078*** (3.02)	0.067** (2.44)	0.066** (2.44)	0.038 (1.40)
fin	0.001 (0.06)	0.006 (0.45)	-0.000 (-0.03)	0.001 (0.05)	0.002 (0.19)

_cons	0.270*** (15.56)	0.274*** (13.45)	0.310*** (12.22)	0.263*** (9.26)	0.275*** (14.83)
Countries	60	60	60	60	60
N	1051	1298	1033	1051	973
RMSE	0.247	0.249	0.247	0.247	0.247

t-Statistics are reported in parentheses. *, ** and *** indicate the rejection of the null hypothesis at the 10%, 5% and 1% statistical significance levels, respectively. RMSE denotes squared root of mean squared error.

Table 4: Saving Regression. Total Sample.

	(1) Savings	(2) Savings	(3) Savings	(4) Savings	(5) Savings
wage_share	−0.282*** (−7.39)		−0.288*** (−7.56)	−0.391*** (−2.91)	−0.421*** (−7.06)
gini		0.012 (0.22)	−0.117** (−2.15)		
wage_share ²				0.126 (0.85)	
wage_share(−1)					0.065 (1.05)
wage_share (−2)					0.128** (2.23)
y_gap	0.000 (0.73)	0.001** (2.19)	0.000 (0.67)	0.000 (0.75)	0.000 (0.43)
k_openness	−0.007** (−2.12)	−0.007** (−2.20)	−0.008** (−2.38)	−0.007** (−2.03)	−0.007* (−1.92)
rel_pcgdp(−1)	0.012 (0.76)	−0.019 (−0.98)	−0.008 (−0.46)	0.012 (0.77)	0.013 (0.85)
rel_pcgdp(−1)*k_open- ness	0.012** (2.27)	0.012** (2.04)	0.016*** (2.79)	0.011** (2.18)	0.013** (2.35)
5y_gdp_f	0.255* (1.95)	0.326*** (2.89)	0.260* (1.95)	0.257** (1.96)	0.280* (1.90)
nfa(−1)	0.019*** (2.75)	0.016** (2.17)	0.022*** (3.07)	0.019*** (2.73)	0.020*** (2.91)
nfa_60(−1)	−0.009 (−1.27)	−0.005 (−0.71)	−0.010 (−1.49)	−0.009 (−1.26)	−0.011 (−1.64)
dep_young	−0.371*** (−10.79)	−0.197*** (−5.95)	−0.376*** (−11.04)	−0.383*** (−10.28)	−0.379*** (−10.77)
dep_elderly	−0.457*** (−5.03)	−0.291*** (−2.96)	−0.526*** (−5.57)	−0.465*** (−5.10)	−0.514*** (−5.59)
cred_tot	−0.001 (−0.11)	−0.003 (−0.44)	0.002 (0.24)	−0.001 (−0.09)	−0.002 (−0.32)
fiscal_balance	0.215*** (7.59)	0.211*** (7.00)	0.217*** (7.52)	0.213*** (7.49)	0.189*** (6.60)
fin	0.054*** (3.32)	0.067*** (3.83)	0.053*** (3.33)	0.054*** (3.33)	0.055*** (3.34)
_cons	0.349*** (16.47)	0.234*** (9.10)	0.401*** (12.99)	0.372*** (10.94)	0.320*** (14.11)
Countries	60	60	60	60	60
N	1051	1298	1033	1051	973
RMSE	0.243	0.241	0.244	0.243	0.244

t-Statistics are reported in parentheses. *, ** and *** indicate the rejection of the null hypothesis at the 10%, 5% and 1% statistical significance levels, respectively. RMSE denotes squared root of mean squared error.

When we include the wage share and the Gini coefficient together, the last one is significant and negative both in investment and saving regressions. This is consistent with the result that the Gini is neutral to the current account, but it represents some challenges to the interpretation of the relationship between inequality and saving. In fact, both inequality indicators have significant and negative coefficients, but they are supposed to be negatively correlated. This reveals the relevance of being precise when we talk about “*inequality*” and emphasizes the need of a more comprehensive analysis of how different inequality indicators are related.

For the control variables, we found some differences between the determinants of investment and saving. The output gap, the forecasted growth and the credit/GDP ratio are more relevant to investment, while the net foreign assets, the capital account openness and the fiscal balance are more relevant in the saving regressions. The dependency ratios seem to be highly relevant even for investment and for saving.

Considering the structural differences between advanced and emerging economies, we tested the validity of the previous results by dividing the sample into these two groups.

Table 5 shows⁸ that the wage share is significant and with negative sign in the current account and in the saving regressions for each group of countries, and not significant in the investment regression. However, for emerging countries this coefficient more than doubles the one in advanced countries. Regarding the Gini coefficient, we find it significant in investment and saving regressions only for the group of advanced countries. This result justifies our decision to focus on the functional distribution of income when we use a sample including non-advanced economies.

Table 5: Advanced vs. Emerging Countries.

	Current account		Investment		Savings	
	Adv	Eme	Adv	Eme	Adv	Eme
wage_share	−0.121** (−2.00)	−0.270*** (−5.84)	−0.016 (−0.42)	−0.042 (−0.87)	−0.121** (−2.23)	−0.334*** (−6.31)
gini	−0.048 (−0.64)	−0.032 (−0.47)	−0.103** (−2.17)	−0.072 (−1.00)	−0.131* (−1.94)	−0.123 (−1.54)
y_gap	−0.003*** (−5.21)	−0.003*** (−4.40)	0.004*** (8.65)	0.003*** (5.07)	0.001 (1.30)	−0.000 (−0.03)
k_opennes	−0.031* (−1.66)	0.003 (0.55)	−0.001 (−0.10)	−0.015** (−2.41)	−0.039** (−2.34)	−0.012* (−1.84)
rel_pcgdp(−1)	−0.035 (−0.82)	−0.029 (−0.54)	−0.030 (−1.01)	−0.079 (−1.39)	−0.078** (−2.01)	−0.102 (−1.60)
rel_pcgdp(−1)*k_opennes	0.043** (2.29)	−0.033 (−1.32)	−0.007 (−0.53)	0.069*** (2.90)	0.043** (2.57)	0.040 (1.59)
5y_gdp_f	0.043 (0.21)	−0.656*** (−3.09)	0.218 (1.44)	1.045*** (5.37)	0.146 (0.79)	0.398** (1.96)
nfa(−1)	0.025*** (3.82)	0.032 (1.59)	0.003 (0.62)	−0.003 (−0.18)	0.028*** (4.83)	0.008 (0.43)
nfa_60(−1)	−0.029*** (−4.17)	−0.036** (−2.30)	0.013*** (2.65)	0.027* (1.84)	−0.016*** (−2.65)	0.001 (0.06)
dep_young	−0.014 (−0.16)	−0.284*** (−6.14)	−0.322*** (−5.94)	−0.199*** (−4.08)	−0.367*** (−4.56)	−0.489*** (−9.00)
dep_elderly	0.080 (0.76)	−0.661*** (−3.46)	−0.441*** (−6.91)	−0.386* (−1.90)	−0.388*** (−4.15)	−1.053*** (−4.64)
cred_tot	−0.021*** (−2.89)	−0.002 (−0.13)	0.015*** (3.21)	0.020 (1.48)	−0.008 (−1.20)	0.016 (1.07)
fiscal_balance	0.223*** (5.11)	0.173*** (3.52)	0.178*** (5.67)	0.007 (0.15)	0.377*** (9.61)	0.161*** (3.62)
fin	0.052*** (4.44)		−0.002 (−0.26)		0.048*** (4.55)	
_cons	0.078 (1.25)	0.100** (2.27)	0.320*** (7.54)	0.303*** (6.59)	0.395*** (7.05)	0.410*** (8.02)
Countries	25	35	25	35	25	35
N	580	453	580	453	580	453
RMSE	0.045	0.047	0.241	0.255	0.250	0.239

t—Statistics are reported in parentheses. *, ** and *** indicate the rejection of the null hypothesis at the 10%, 5% and 1% statistical significance levels, respectively. RMSE denotes squared root of mean squared error.

The other variables behave similarly to before, but there are some relevant differences between each group of countries. The forecasted growth is highly significant and positive for saving and investment in emerging countries, resulting in a negative relation with the current account, while the net foreign assets are only significant and positive in the saving and current account regressions of advanced economies. As could be expected, credit to GDP is important and negative for the current account mainly because of its impact on investment in advanced economies. Fiscal balance is positive and significant for saving in both advanced and emerging countries. But it is relevant and positive only for investment in advanced economies.

The interaction between GDP per capita and capital account openness is significant in emerging countries only on the investment side. According to the estimated coefficients, in around two-thirds of the emerging

countries included in our sample⁹ the openness of the capital account is related to a decrease in investment. On the other hand, this interaction is significant in advanced economies only on the saving side. The openness of the capital account is related to an increase in savings in around two-thirds of the advanced economies considered.¹⁰ In the advanced economies, a higher GDP per capita is always related with a decrease in savings.¹¹

As we mentioned in Section 5, many cross-country panel studies use fixed effects in order to soak up all country-specific effects. We think that including fixed effects would detract from much of the economically meaningful parts of the analysis, and this is the reason why most earlier studies (Chinn and Prasad 2003; Chinn and Ito 2007; Chinn, Eichengreen, and Ito 2012; Phillips et al. 2014) do not include fixed effects in their main regressions. However, in order to test the robustness of our results we include the main regression with country-fixed effects. In the first three columns of Table 6 we can see that the wage share remains highly significant and with a negative sign in the current account and saving regressions, while it is not significant in the investment regression. The other variables perform in a similar way to our previous results.

Table 6: Regression with Fixed Effects and Controlling by REER.

	c_Corriente	Invest	Fixed effects Savings	c_Corriente	Controlling by REER Invest	Controlling by REER Savings
wage_shere	-0.279*** (-4.74)	0.014 (0.25)	-0.265*** (-5.00)	-0.237*** (-6.43)	-0.019 (-0.60)	-0.275*** (-7.04)
gini	-0.046 (-0.82)	-0.084 (-1.60)	-0.130** (-2.55)	-0.088 (-1.55)	-0.032 (-0.66)	-0.096 (-1.58)
reer(-1)				-0.041*** (-3.25)	0.041*** (3.75)	-0.005 (-0.43)
reer(-2)				0.009 (0.90)	0.010 (1.17)	0.016* (1.86)
y_gap	-0.003*** (-7.48)	0.003*** (8.61)	0.000 (0.54)	-0.003*** (-5.61)	0.003*** (7.09)	0.000 (0.87)
k_opennes	-0.005 (-1.38)	0.010*** (3.21)	0.005* (1.76)	-0.001 (-0.21)	-0.007** (-2.03)	-0.005 (-1.32)
rel_pcgdp(-1)	0.037* (1.73)	0.038* (1.92)	0.076*** (3.88)	0.021 (1.24)	-0.039*** (-2.74)	-0.010 (-0.57)
rel_pcgdp(-1)*k_opennes	0.010** (2.17)	-0.017*** (-3.92)	-0.007 (-1.62)	0.010 (1.61)	0.005 (0.86)	0.012** (2.04)
5y_gdp_f	-1.022*** (-6.74)	0.972*** (6.92)	-0.050 (-0.37)	-0.365** (-2.21)	0.768*** (5.34)	0.311** (2.03)
nfa(-1)	0.029*** (4.44)	-0.016*** (-2.68)	0.013** (2.17)	0.023*** (3.04)	0.005 (0.69)	0.017** (2.31)
nfa_60(-1)	-0.034*** (-5.44)	0.040*** (6.80)	0.005 (0.94)	-0.019** (-2.52)	0.010 (1.43)	-0.003 (-0.46)
dep_young	-0.064 (-1.56)	-0.107*** (-2.80)	-0.172*** (-4.60)	-0.063 (-1.52)	-0.215*** (-6.04)	-0.284*** (-6.28)
dep_elderly	-0.235*** (-2.65)	-0.300*** (-3.66)	-0.535*** (-6.70)	-0.172* (-1.82)	-0.176** (-2.19)	-0.361*** (-3.52)
cred_tot	-0.044*** (-7.54)	0.018*** (3.42)	-0.025*** (-4.86)	-0.006 (-0.88)	0.010 (1.59)	0.001 (0.16)
fiscal_balance	0.150*** (4.24)	0.176*** (5.38)	0.326*** (10.22)	0.276*** (6.62)	0.112*** (3.10)	0.350*** (9.11)
fin				0.051*** (3.80)	0.004 (0.33)	0.063*** (4.21)
_cons	0.087** (2.21)	0.245*** (6.74)	0.332*** (9.36)	0.132*** (3.96)	0.244*** (8.52)	0.375*** (10.68)
Countries	60	60	60	47	47	47
N	1033	1033	1033	837	837	837
RMSE	0.044	0.247	0.242	0.045	0.244	0.244

t-Statistics are reported in parentheses. *, ** and *** indicate the rejection of the null hypothesis at the 10%, 5% and 1% statistical significance levels, respectively. RMSE denotes squared root of mean squared error.

We also want to check if the negative relationship between the wage share and the current account is due to an income effect or an exchange rate effect, as we discussed in Section 4. Considering that the current account and the real effective exchange rate (REER) are both endogenous and simultaneously determined, we included two lags of the REER in the regressions aiming to capture the presence of the exchange rate effect. As we can see in the last three columns of Table 6, the first lag of the REER is significant in the current account and investment

regressions. However, the significance and size of the coefficient of the wage share are roughly the same as the ones in Table 2, Table 3 and Table 4. We interpret this result as evidence of the presence of both income and exchange rate effects, although further analysis is required. The number of countries with REER data is a bit lower; however, the comparison shows very similar results in sign and coefficients for most of the control variables.

All of these different approaches performed in our analysis confirm the main hypothesis of this paper, that the wage share is an important determinant of the current account. A higher wage share corresponds to a lower current account balance. Even though more analysis is needed in order to be more conclusive, the results of the separate regressions of saving and investment and the control by the presence of the exchange rate effects support the idea that there is an income effect: a higher wage share is associated with higher consumption, less saving, and a decrease in the current account, but is not associated with a decrease in total investment.

7 Conclusions

The increase in income inequality in recent decades is a phenomenon that involves almost all countries, both advanced and emerging ones. In this context the economic literature has shown a renewed interest in distribution issues, although with a strong predominance of advanced countries in the empirical research. Our use of the functional distribution of income, rather than other data sets that have predominated in recent studies, has allowed us to extend our analysis to a larger, more inclusive group of countries.

As shown by the results of this paper, the share of wages in total income is a significant determinant of the external accounts: the higher the wage/GDP ratio, the lower the current account. This result is robust for different specifications of the model and different panel compositions. We consider it an improvement on other works about the determinants of the current account (Chinn and Prasad 2003; Chinn and Ito 2007; Chinn, Eichengreen, and Ito 2012; Phillips et al. 2014) because we are proving the importance of an additional and very significant covariate. This also complements recent literature that connects inequality with the current account, but by using top income distribution (Al-Hussami and Remesal 2012; Behringer and van Treeck 2013; Kumhof et al. 2012).

Our result is compatible with the post-Keynesian literature that suggests a negative relationship between the wage share and the trade balance due to a cost or exchange rate effect that reduces the competitiveness of exports (Blecker 1989; Stockhammer and Wildauer 2016). However, after controlling with the REER, the negative relationship between the wage share and the current account did not disappear, suggesting the presence of an income effect. Further research is required in order to fully disentangle this result.

The wage share is negatively related with saving, which confirms the result of the literature that the wage share has a positive relationship with consumption (Hein and Vogel 2008; Stockhammer, Onaran, and Ederer 2009; Blecker 2016). We did not find any significant relationship between the wage share and total investment, but this could be explained by the different role of business and residential investment. The rest of the variables behave toward saving and investment as expected, in a similar fashion to that described in previous literature.

By dividing the sample between advanced and emerging countries, we observe that the negative relationship between wage share and the current account weakens in the first group and strengthens in the less developed economies. This may be due to the widening of the wage gap, indicating that in advanced economies there is a group of wage earners with a high capacity for saving, whose marginal propensity to consume is more similar to the level usually attributed to those who derive their income from rents and capital. Alternatively, this could be result of the presence of consumption fueled by debt, as is suggested in the recent literature.

At the same time, our paper brings to the surface the importance of structural differences between advanced and emerging economies. The differences in significance and even in direction found in key variables such as the credit/GDP ratio, GDP per capita, the growth expectation, and the demographic ratios, reveal, in turn, the differences between the two groups of countries. Indeed, credit, as a representation of financial intermediation, is a significant determinant of the current account only in advanced countries. Regarding the GDP growth expectations variable, it is not significant for advanced economies, but it is highly relevant for emerging economies.

Our results contribute to different strands of the literature. First, the fact that income distribution, and specifically the wage share, has significant repercussions on the current account, must be considered more accurately when building and testing macroeconomic models. Second, the results illustrate the importance of being precise when analyzing inequality because of the different results expected for wage share vis-à-vis top income or the Gini coefficient. Third, there is a clear differentiation between the role of saving and investment channels in these results. Even though the wage share could be related with different kinds of investment, its relationship with the current account seems to be mostly through the saving channel. Fourth, capital account

openness and financial sector intermediation seem to be very important and to play a different role according to the stage of a country's development. Finally, any domestic economic policy that seeks to simultaneously succeed in reaching an external target and an improvement in income distribution should take this important interaction into account.

Done!

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A Appendix

Table 7: Correlations Between Wage Share and Income Captured by the Top 1%.

Country	Correlation
Argentina	−0.735
Australia	−0.785
Canada	−0.824
China	−0.819
Colombia	0.237
Denmark	0.395
Finland	−0.729
France	−0.434
Germany	−0.781
Ireland	−0.007
Italy	−0.907
Japan	−0.923
Netherlands	−0.477
New Zealand	−0.697
Norway	−0.634
Portugal	0.764
Republic of Korea	0.546
Singapore	0.118
South Africa	−0.665
Spain	−0.407
Sweden	−0.784
Switzerland	−0.445
United Kingdom	−0.013
United States	−0.256

Table 8: Correlations between Wage Share and Gini Coefficient.

Country	Correlation	Country	Correlation
Argentina	−0.644	Mexico	0.852
Australia	−0.844	Mongolia	−0.545
Austria	−0.304	Morocco	0.760
Belgium	−0.537	Netherlands	−0.335
Bolivia	0.049	New Zealand	−0.735
Brazil	−0.097	Niger	−0.208
Burkina Faso	0.880	Nigeria	−0.427
Canada	−0.725	Norway	−0.312
Chile	0.771	Panama	0.913
China	−0.733	Papua New Guinea	−0.896
Colombia	−0.257	Peru	0.836
Costa Rica	0.777	Philippines	0.619
Cyprus	0.322	Poland	−0.425
Denmark	0.396	Portugal	−0.385
Egypt	−0.170	Republic of Korea	−0.929
Finland	−0.657	Romania	0.537
France	0.584	Senegal	−0.563
Germany	−0.713	Sierra Leone	−0.344
Greece	−0.369	Singapore	−0.207
Hungary	−0.368	South Africa	0.067
Iceland	0.720	Spain	0.452
Iran	0.469	Sri Lanka	0.587
Ireland	−0.770	Sweden	−0.770
Israel	−0.678	Switzerland	−0.389
Italia	−0.891	Thailand	−0.446
Ivory Coast	−0.602	Tunisia	−0.087
Japan	−0.893	Turkey	−0.007
Kenya	−0.437	United Kingdom	−0.160
Lesotho	0.689	United States	−0.512
Luxembourg	−0.219	Venezuela	−0.677

B Appendix

Correlation between Functional and Personal Income Distribution

The correlation between the wage share and the indicators of personal distribution of income remain stable both when contemporary values of each indicator are considered and when a lag or lead in any of the variables is included. This is observed in the Figure 1A and 2B.¹²

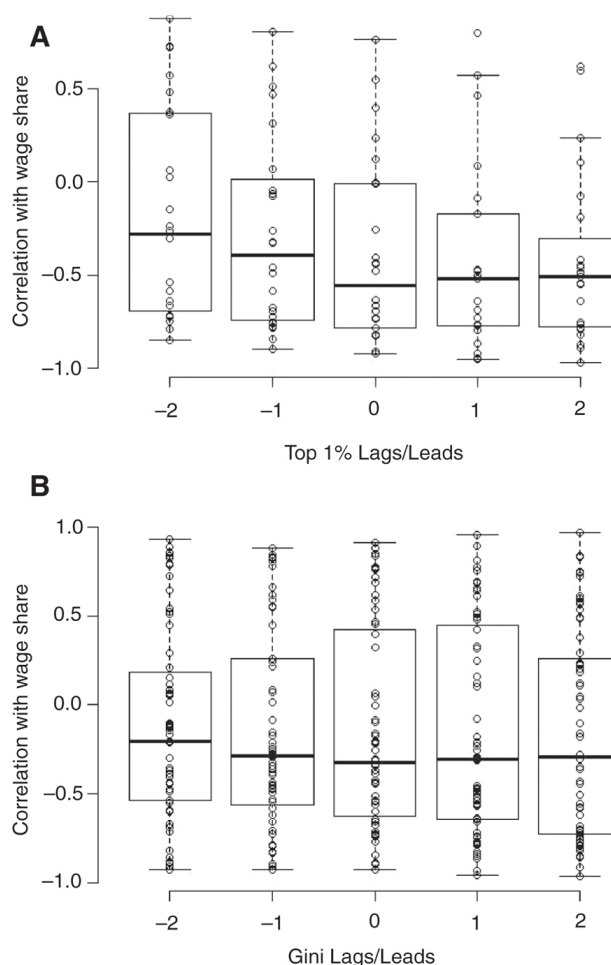


Figure 2: A. Boxplot for the Correlations of Wage Share and Top 1% Lags/Leads. B. Boxplot for the Correlations of Wage Share and Gini

It may be observed that the sample mean adopts negative values in all cases. Moreover, the median is always below the mean, and this suggests that there is a higher concentration of correlations in the negative tail. This is confirmed when noting that the outliers, if any, are positive, and the lower limit of the boxes always adopts a value equal to or lower than -0.5 , while the upper limit rarely comes close to the 0.5 value and even adopts negative values in some cases.

C Appendix

Sources of the variables

Current Account (% GDP) ("CA"). Lane and Milesi-Ferretti (2007), updated to 2011. IMF's World Economic Outlook (WEO) database for Belgium (1980–1994) and Hungary (1980–1981).

Investment (%GDP) ("invest"). World Bank's WDI database.

Wage Share in Total Income ("wage_share"). Karabarounis and Neiman (2013).

Gini Coefficient ("gini"). "Standardized World Income Inequality Database," compiled by Solt (2016), versions 4.0 and 5.0.

Net Foreign Assets Position (% GDP) ("nfa"). Lane and Milesi-Ferretti (2007). World Bank's WDI database for China (1977–1980), Luxembourg (1983–1998), Mongolia (1991), and Romania (1981–1989).

Population Dependence Ratios ("dep_young" and "dep_elderly"). World Bank's WDI database.

Credit to the Private Sector (% GDP) ("cred_tot"). World Bank's "Financial Development and Structure Dataset," and the measure comprises the sum of private credit from banks and from other financial institutions. The series for Chile and Venezuela have been completed with data on "Domestic Credit to the Private Sector" from the World Bank's World Development Indicators (WDI) database.

Fiscal Balance (% GDP) ("fiscal_balance"). IMF's WEO database (general government net lending/borrowing), supplemented with data from the OECD (net lending/net borrowing) for Australia (1975–1987), Finland

(1975–1979), Israel (1995–1999), Korea (1975–1994), the U.S. (1975–2000), and South Africa (1995–1999). We also use data from the European Commission's "Annual Macro-Economic Database" (AMECO) (net lending/net borrowing general government) for Canada (1975–1979), France (1975–1979), Romania (1995–1999) and Turkey (2000–2001).

Output Gap (% GDP) ("y_gap"). For 25 countries we use data from the IMF World Economic Outlook Database. For the rest of the sample we estimated with a Hodrick-Prescott filter.

Five-Year Growth Forecast ("5y_gdp_f"). IMF's WEO database. As these series start in 1988, for earlier years we use as a proxy the average real growth of the previous 5 years, employing economic growth data from the World Bank's WDIs.

PPP-converted GDP Per Capita ("rel_pcgdp"). Penn World Table 7.1 up to 2010. For 2011, values are estimated on the basis of relative GDP growth and population growth (data from the World Bank's WDI database).

Capital Account Openness Index ("k_openness"). Chinn and Ito (2006), online version updated to 2013. Switzerland and Luxembourg series were completed with the maximum levels of openness possible.

Real Effective Exchange Rate ("reer"). IMF International Financial Statistics Database.

D Appendix

Countries Included in the Sample

Africa (12): Burkina Faso, Egypt, Ivory Coast, Kenya, Lesotho, Morocco, Niger, Nigeria, Senegal, Sierra Leone, South Africa and Tunisia.

America (12): Argentina, Bolivia, Brazil, Canada, Chile, Colombia, Costa Rica, Mexico, Panama, Peru, the U.S. and Venezuela.

Asia (10): China, Iran, Israel, Japan, Mongolia, the Philippines, the Republic of Korea, Singapore, Sri Lanka and Thailand.

Europe (23): Austria, Belgium, Cyprus, Denmark, Finland, France, Germany, Greece, Hungary, Iceland, Ireland, Italy, Luxembourg, Norway, the Netherlands, Poland, Portugal, Romania, Spain, Sweden, Switzerland, the United Kingdom (U.K.) and Turkey.

Oceania (3): Australia, New Zealand and Papua New Guinea.

According to the classification used by the IMF, the partition between advanced and emerging countries is as follows:

Advanced Countries (25): Australia, Austria, Belgium, Canada, Denmark, Finland, France, Germany, Greece, Iceland, Ireland, Israel, Italy, Japan, Luxembourg, the Netherlands, New Zealand, Norway, Portugal, Singapore, Spain, Sweden, Switzerland, the U.K. and the U.S.

Emerging Countries (35): Argentina, Bolivia, Brazil, Burkina Faso, Chile, China, Colombia, Costa Rica, Cyprus, Egypt, Hungary, Iran, Ivory Coast, Kenya, Lesotho, Mexico, Mongolia, Morocco, Niger, Nigeria, Panama, Papua New Guinea, Peru, the Philippines, Poland, the Republic of Korea, Romania, Senegal, Sierra Leone, Sri Lanka, South Africa, Thailand, Tunisia, Turkey and Venezuela.

Notes

¹"[A] declining labour income share can limit household consumption and reduce overall aggregate demand if the redistribution of income to capital does not sufficiently increase investment or if lower wages do not increase net exports sufficiently to offset lower domestic demand. These negative consumption effects can in turn weaken investment, as firms do not see new strong sources of demand. The resulting negative effect on global demand may limit exports and reduce overall economic growth." (ILO, IMF and OECD 2015). Prepared for the G20 Labour and Employment Ministers Meeting and Joint Meeting with the G20 Finance Ministers, Ankara, Turkey, 3–4 September 2015.

²The current account can be seen as the excess of savings over investment ($CA = S - I$). If an increase in consumption means a decrease in savings without affecting the level of investment, there is a reduction in the current account.

³For simplicity, in this paper we have considered as "emerging countries" both emerging and developing Economies.

⁴We used data from "The World Top Incomes Database".

⁵We used data from the "Standardized World Income Inequality Database," prepared by Solt (2016) on the basis of data from different sources compiled by the Luxembourg Income Study (LIS).

⁶For the ratio savings/GDP we use (current account+investment)/GDP.

⁷As in Al-Hussami and Remesal (2012), Behringer and van Treeck (2013), and Phillips et al. (2014), the demeaning was undertaken using the following formula:

$$\tilde{X}_{it} \equiv X_{it} - \frac{\sum_{i=1}^I PIB_{it} X_{it}}{\sum_{i=1}^I PIB_{it}}$$

where X_{it} is the original variable, $\bar{X}_{it}$ is the demeaned variable, PIB_{it} is GDP at current values taken from the *External Wealth of Nations* dataset of Lane and Milesi-Ferretti (2007), subindex i refers to the country and t to the year.

⁸We only present the results for the third model. The regressions for the rest of the models are available upon request.

⁹Twenty-two countries out of 35 have an average relative GDP per capita lower than the 0.22 required to turn positive the partial derivative $\frac{d(\text{invest})}{d(k_openness)} = -0.015 + 0.069 \times \text{rel_pcgdp}$

¹⁰Sixteen countries of 25 have an average relative GDP per capita higher than the 0.91 required to turn positive the partial derivative $\frac{d(\text{savings})}{d(k_openness)} = -0.039 + 0.043 \times \text{rel_pcgdp}$

¹¹For any considered country, the sign of the partial derivative $\frac{d(\text{savings})}{d(\text{rel_pcgdp})} = -0.078 + 0.043 \times k_openness$ is negative.

¹²The “+” sign represents the sample mean, while the thick horizontal line represents the median. The “boxes” represent the second and third quartiles of the distribution, from 25% to 75%, i.e. they contain the central 50% of the sample. The height of the box represents the interquartile range (IQR), while the whiskers represent the last value of the sample between the respective quartile and a 1.5 IQR distance. The white circles represent the estimated correlations for each country; when they are out of the reach of the whiskers, they are considered outliers.

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